

Unit 9, Lesson 5: Cavalieri's Principle



Benchmark

MA.912.GR.4.5 Solve mathematical and real-world problems involving the volume of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.



Learning Target

- ☐ I can use Cavalieri's principle to give informal arguments about the formulas for the volumes of right and non-right cylinders, pyramids, prisms, and cones.



9.5.1 Warm-Up: Notice and Wonder

Four different perspectives of the same stack of records are shown.



1. What do you notice?
2. What do you wonder?



9.5.2 Collaboration: Exploring Cavalieri's Principle

Scenario 1

Consider the two stacks of pennies, where each stack contains the same number of pennies. Discuss with your partner the relationship between the volumes of each stack.

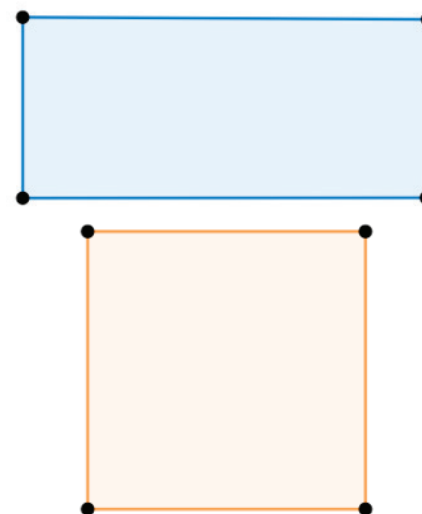


1. Summarize your discussion in the table below and have your partner initial that your summary is correct.

Summary	Partner's Initials

Scenario 2

Suppose two partners are both constructing towers that are to be 5 inches high. Partner A decided to build a tower using rectangles with areas of 16 square inches and partner B decided to construct a tower using squares with areas of 16 square inches. The base of each partner's tower is shown.

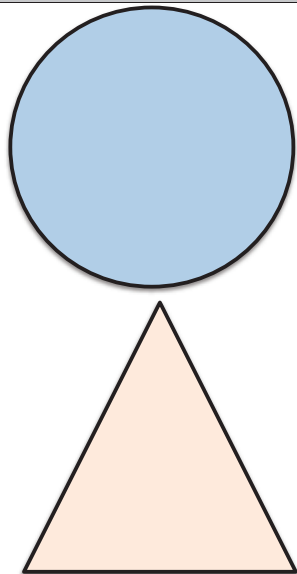


2. Discuss with your partner how the volumes will compare. Summarize your findings in the box below and have your partner initial that your summary is correct.

Summary	Partner's Initials

Scenario 3

ABC ice cream shop decided to run a special for the week and they're offering two serving options for you to choose from. Option 1 is a cone with a circular base that has an area of 4π square inches and a height of 4 inches. Option 2 is a triangular pyramid container with a base of 4π square inches and a height of 4 inches. The base of each option is shown.



3. Discuss with your partner how the volumes will compare to one another. Summarize your findings in the box below and have your partner initial that your summary is correct.

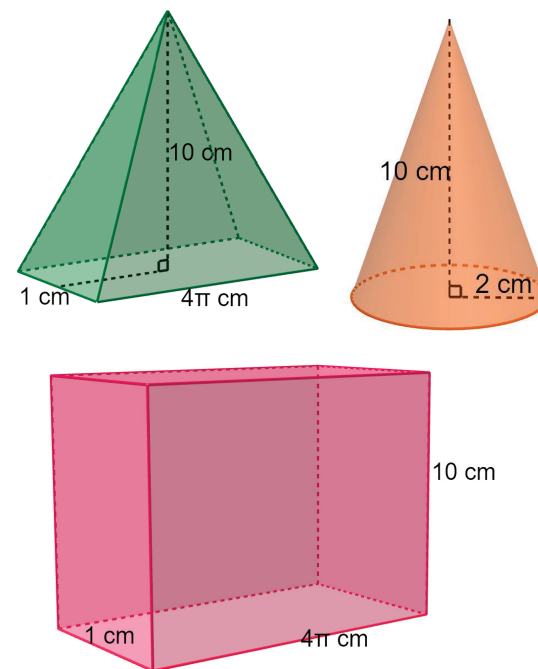
Summary	Partner's Initials

9.5.3 Guided Instruction: Cavalieri's Principle

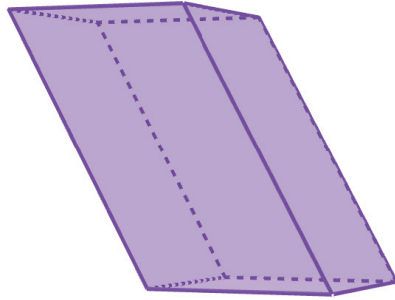


If every plane parallel to the two bases of two solids results in cross sections of equal area and the two solids have congruent altitudes, then the solids have equal volumes.

1. Using Cavalieri's Principle, determine whether or not the following solids have the same volumes.



2. An oblique prism is shown.

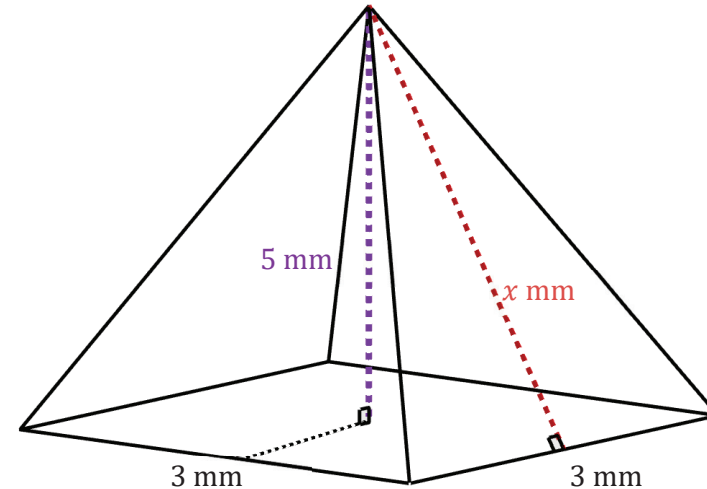


- What shape is each cross-section of the oblique prism?
- What is the area formula for the base?
- What is the volume formula for an oblique rectangular prism?
- Discuss with your partner how the formula for the oblique rectangular prism compares to the formula for a right rectangular prism. Explain your reasoning.



9.5.4 Guided Instruction: Slant Height vs. Vertical Height

A square pyramid is shown with measurements in millimeters (mm).



Samantha and Julian each found the measure of the slant height, x , by first drawing a diagram and then applying the Pythagorean Theorem. Their work is shown.

Samantha	Julian
$c^2 = 3^2 + 5^2$	$c^2 = 1.5^2 + 5^2$

1. Discuss with your partner which student correctly set up their work to determine the slant height. Summarize your discussion.



The **slant height** of an object is the distance from the base to the apex along a lateral face, whereas the **vertical height or altitude** of an object is the perpendicular distance from the apex to the base.

2. Discuss with your partner which height is important when finding the volume of an oblique figure. Summarize your discussion.



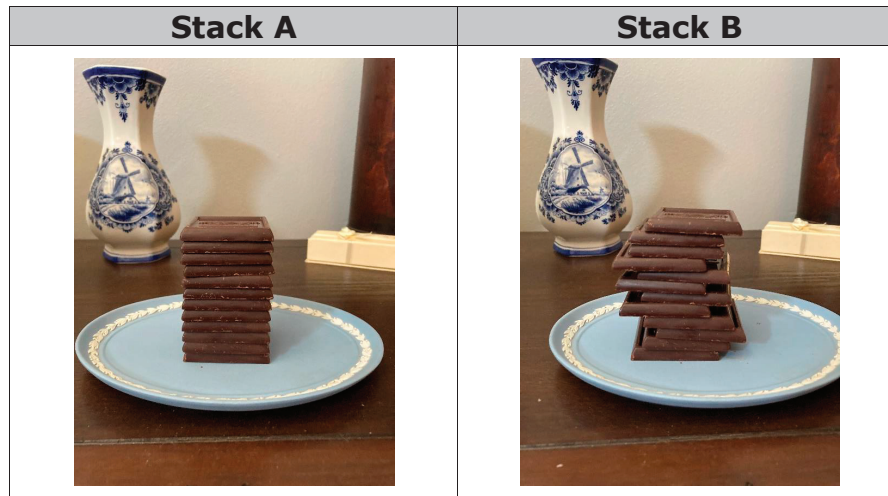
9.5.5 Guided Practice: Cavalieri's Principle in the Real World

1. Consider the stack of sticky note pads below.



If each pad is a square with side lengths of 3 inches and a height of 0.97 inches, approximate the volume of a stack of 8 sticky note pads.

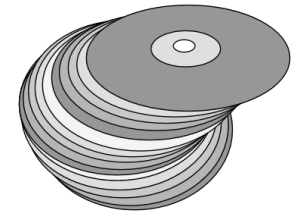
2. Two stacks of 12 chocolate squares are shown.



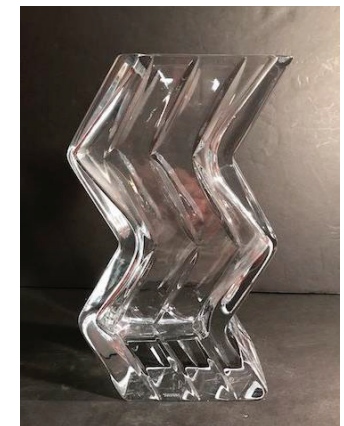
Each chocolate square has the dimensions $1\frac{5}{8}$ inches (in.) by $1\frac{5}{8}$ in. Each square is $\frac{1}{8}$ in. thick. Find the volume of each stack.

Stack A	Stack B

3. While cleaning out an attic, you found an old stack of compact discs (CDs) that were used to store music. If each disc has an area of 36π cm², explain how you can find the approximate volume of a stack of 25 CDs.



4. A zigzag crystal vase is shown, where the vase has a height of 10 inches (in.). The cross-sections that are parallel to the base are always rectangles that are 6 in. wide by 3 in. long.



- Determine the volume of the vase, assuming the crystal itself has no thickness.
- The crystal is actually half an inch thick on each of the sides and on the bottom. Approximately how much space is contained within the vase? Explain or show your reasoning.



9.5.6 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Unit 9, Lesson 6: Finding Volume



Benchmark

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Learning Target

- ☐ I can find the volume of spheres, prisms, cones, pyramids, and cylinders.